

LECTURE: INTRODUCTION TO RAG

Retrieval-Augmented Generation (RAG) is an architectural pattern that extends the capabilities of language models by using an external knowledge source.

1. Indexing

At this stage, documents are split into chunks. For each chunk, an embedding is calculated - a vector representation of the text. Embeddings are stored in a vector store together with the original text.

2. Search

When the user asks a question, an embedding is also calculated for it. The system searches the store for those chunks most similar to the question by cosine similarity.

3. Generation

The found chunks are fed to the LLM as context. The model formulates an answer based on this context, not based on its internal knowledge.

Advantages of RAG:

- Reduces hallucinations: the model answers only based on real documents.
- Allows updating knowledge without retraining the model.
- Gives the user sources to verify the answer.